A389475
Two cubic congruences in characteristic three

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Abstract

For the unique normalized integer solution of $V(3x^2) = 2x^2 + V^2 - 2V^3$, we prove both recorded modulus-three identities. The construction is integral and unique; the congruence argument is a reduction of the source equation followed by standard Frobenius.

1 Source and conclusions

The A389475 series [1] is the normalized integer solution

$$V(3x^2) = 2x^2 + V(x)^2 - 2V(x)^3, \qquad V = x + O(x^2).$$

The offset is one. We prove the two coefficientwise identities

$$V(x)^3 \equiv V(x^3) \pmod 3, \qquad V(x)^2 \equiv x^2 + 2V(x^3) \pmod 3.$$

The first is standard Frobenius; the second uses this source equation.

2 Integral construction and uniqueness

Theorem 2.1. *For every $s \in \mathbb{Z}$, there is a unique normalized series $R_s \in \mathbb{Z}[[x]]$ satisfying*

$$R_s(3x^2) = 2x^2 + R_s(x)^2 + 2sR_s(x)^3. \tag{1}$$

Proof. For any normalized integral series U, define its residual by

$$E_s(U) = U^2 + 2sU^3 - U(3x^2) + 2x^2.$$

Every coefficient of $E_s(U)$ is even. Indeed, after reduction modulo two, Frobenius gives $U(x)^2 = U(x^2)$ and $3x^2 = x^2$, so the residual vanishes. Also $E_s(U) = O(x^3)$ for every normalized U.

Suppose $V - U = cx^r + O(x^{r+1})$, with $r \geq 2$. The quadratic difference has leading term $2cx^{r+1}$; the cubic difference has order at least $r + 2$; and $(V - U)(3x^2)$ has order at least $2r \geq r + 2$. Consequently

$$E_s(V) - E_s(U) = 2cx^{r+1} + O(x^{r+2}). \tag{2}$$

Starting with $U = x$, choose its degree-r coefficient to cancel the degree-$(r+1)$ coefficient of the residual. The needed division by two is exact by the preceding parity argument. Equation (2) shows that earlier cancellations remain valid. Each degree is thus determined by finitely many earlier coefficients, and the resulting integral series has zero residual in every degree.

For uniqueness, take the first degree $r \geq 2$ where two candidate solutions differ. Equation (2) would give a nonzero degree-$(r+1)$ difference of residuals, because $2c \neq 0$ in $\mathbb{Z}$. This contradicts both residuals being zero. □

The proof formalizes divisibility of the residual by two before using integer division. Thus the construction does not assume the existence or integrality of an OEIS sequence. The names used below denote the uniquely constructed series.

Taking $s = -1$ in Theorem 2.1 constructs the desired series $V = R_{-1}$ and proves uniqueness for its literal source equation.

3 The standard Frobenius identity

For any integer formal series W and prime p, coefficient reduction gives

$$\overline{W}(x)^p = \overline{W}(x^p) \qquad \text{in } \mathbb{F}_p[[x]].$$

For each fixed coefficient one may truncate W at that degree. The polynomial identity $(u+v)^p = u^p + v^p$ in characteristic p, together with $a^p = a$ in $\mathbb{F}_p$, proves the displayed identity coefficient by coefficient. Thus it holds for every formal series, without a convergence assumption. This is standard algebra; its application below is not a new general principle.

4 The two modulus-three identities

Theorem 4.1. *For every coefficient index $n \geq 0$,*

$$[x^n](V(x)^3 - V(x^3)) \equiv 0 \pmod 3,$$
$$[x^n](V(x)^2 - x^2 - 2V(x^3)) \equiv 0 \pmod 3.$$

Proof. Apply Frobenius with $p = 3$ and $W = V$ for the first identity. For the second reduce the source equation coefficientwise to $\mathbb{F}_3$. Since $V(0) = 0$, its left side reduces to $V(0) = 0$. Hence

$$0 = 2x^2 + V^2 - 2V^3, \qquad V^2 = x^2 + 2V^3.$$

Substitute the first identity to obtain $V^2 = x^2 + 2V(x^3)$. Equality of series gives every stated coefficient congruence. □

This proves the assertions labeled (4.a) and (4.b) in the retrieved entry. Both hold for the full series, including degree zero.

5 Lean proof and exact scope

The dedicated entry file is `lean/Entries/A389475.lean`. Its declarations lie in namespace `FinalCollection.A389475`. The principal checked endpoints are:

- `source_equation`: the literal defining equation.
- `source_existence_uniqueness`: the integral source construction and uniqueness.

- `integral_family_existence`: the integer-parameter family used in the construction.
- `cube_mod_three`: the full Frobenius identity.
- `square_mod_three`: the full square identity derived from the source.
- `cube_coefficients`: the first assertion at every coefficient.
- `square_coefficients`: the second assertion at every coefficient.

The full implementations are in `lean/More/CubicConsequences.lean`. Every index is universally quantified. The first congruence is a standard identity valid for every integral formal power series; the second is a consequence of this particular source equation. The companion also separately states normalization and source uniqueness.

The companion is a separate, explicitly typed Lean entry file. Its proofs invoke the preserved implementation modules; the complete dependency source is included in the same archive. Shared lemmas are reused rather than assumed. The entry file is not a self-contained replacement for Mathlib or the supplied common library.

The project pins Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The collection checker recompiles the proof sources and audits the theorems' transitive axiom dependencies. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. The delivered verification reports record the actual successful runs. Finite numerical checks are corroboration; the displayed conclusions are unbounded theorems. An AI review of source correspondence does not replace independent human review.

References

[1] OEIS Foundation, entry A389475, `https://oeis.org/A389475`. Source checked 24 October 2026. The source sequence and selected conjectures are credited to Paul D. Hanna.

[2] D. Goyal, supplied reference repository, `https://github.com/dvgyl/oeis-a396843-proof`. This separate presentation belongs to the final collected proof package; no earlier DOI is reassigned to it.